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Multiple light source configuration

Abstract

A lighting assembly including a plurality of lighting sources or modules arranged concentrically about an inner circumference of the lighting device, wherein the plurality of lighting sources or modules emit light in at least one of a plurality of wavelength ranges (UV, visible (e.g., blue, green, yellow, orange, red, white, etc.), IR) onto a lighting director device that redirects the emitted light toward a lens system, the lens system focusing the light onto a viewing point. Further disclosed is a passthrough within the lighting director device that allows light positioned on a base of the lighting device to pass through the lighting director.

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Background/Summary

CLAIM OF PRIORITY (1) This application claims, pursuant to 35 USC 119, priority to, and the benefit of the earlier filing date of provisional patent application Ser. No. 63/631,327, filed on Apr. 8, 2024, and pursuant to 35 USC 120, as a Continuation-in-Part application, priority to and the benefit of, the earlier filing date of patent application Ser. No. 18/218,503, filed on Jul. 5, 2023, which claimed, pursuant to 35 USC 120, as a Continuation application, priority to, and the benefit of the earlier filing date of patent application Ser. No. 17/744,711, filed on May 15, 2022, which claimed, as a Continuation application, priority to and the benefit of the earlier filing date of patent application Ser. No. 17/527,130, filed on Nov. 15, 2022 (U.S. Pat. No. 11,359,798), which claimed, as a Continuation application, priority to, and the benefit of the earlier filing date of patent application Ser. No. 17/233,467, filed on Apr. 17, 2021, (U.S. Pat. No. 11,231,165), which claimed priority to and the benefit of the earlier filing date, pursuant to 35 USC 119, as a non-provisional application, of patent application Ser. No. 63/013,487, filed on Apr. 21, 2020, the contents of all of which are incorporated by reference, herein.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) This application is related to the field of lighting and more particularly a system for providing uniform light distribution from a plurality of light sources.

Related Application

(2) This application is related to the U.S. Pat. Nos. 7,682,042; 8,851,709; RE 46,463; 9,791,138; 240,769; P1; 10,247,384; 10,527,254, and U.S. patent application Ser. Nos. 16/693,212 and 17/233,543, the contents of all of which are incorporated by reference, herein.

BACKGROUND

(3) Lighting devices are typically used in dental, medical and/or surgical fields to allow practitioners (e.g., dentist, doctor, surgeons, etc.) to apply light directly to the area where the practitioner is viewing. Some lighting devices may be free-standing lamps that a practitioner may position about the work area. Other lighting devices may be overhead lighting devices that operate

on an arm that the practitioner may position about the work area. User-wearable (e.g., head-mounted) lighting devices may also be used by a practitioner to provide a tight light beam directly coincident with the practitioner's line of sight. Head-mounted lighting devices are advantageous as the projected light is directly at the focus of the practitioner's eyes and the practitioner's shadow is not projected onto the work area as in the case of overhead lights.

(4) Operation of such head-mounted devices is known in the art. For example, U.S. Pat. No. 8,851,709; RE46,463; U.S. Pat. Nos. 9,791,138; 10,240,769; and 10,527,254, which are assigned to the Assignee of the instant application, disclose user-wearable (e.g., head mounted) devices and their operation. Similarly, U.S. Pat. No. 7,682,042, which is assigned to the Assignee of the instant application, discloses an overhead or lamp configuration. The contents of all of which are incorporated by reference, herein.

(5) Typically, with a head mounted lighting device, a practitioner (e.g., a dentist, a doctor, a surgeon) adjusts the lighting element such that the light is projected onto a surface to which the practitioner's eyes are focused. The practitioner may then control the light output in a manner as disclosed in the aforementioned US patents.

(6) In addition to projecting a light (such as a white light) for assisting the practitioner in viewing the targeted area, the light source may be composed of different lighting sources that may be used for different purposes. For example, the light sources may generate an ultra-violet light (non-visible wavelength range), a visible light (e.g., white, red, green, blue, etc.) or an Infra-red (IR) (non-visible wavelength), wherein the specific wavelength band may achieve a desired purpose. For example, some light wavelengths are known to decrease the time of gel-like materials to harden. In addition, the lighting devices may comprise one or more of the different light sources may be generating a corresponding light concurrently; the control of which may require wired or wireless control circuitry.

(7) However, user-wearable devices are required to be compact and lightweight, which limits the number light emitting elements that may be used in emitting a light. The limited number of lighting emitting elements limits the overall intensity of the light outputted by the head-mounted lighting device.

(8) Hence, there is a need in the industry for a system that allows for a greater emission distribution of light from lighting devices; particularly of the user-wearable kind.

SUMMARY OF THE INVENTION

(9) In accordance with the principles of the invention, there is disclosed a lighting device for the generation of light from multiple lighting sources that provides a greater emission of the light from lighting devices.

(10) In accordance with the principles of the invention, there is disclosed a lighting device comprising a plurality of lighting sources, wherein each of the lighting sources is configurable to provide light to a desired point or region.

(11) In accordance with the principles of the invention, there is disclosed a lighting device comprising a plurality of lighting sources, wherein the lighting sources may be individually controlled while the light outputted by each of the lighting sources may be directed to a desired point or region

(12) In accordance with the principles of the invention, there is disclosed a lighting device comprising a plurality of lighting sources, wherein the lighting sources may be controlled electronically and physically to direct a light to a desired point or region

(13) In accordance with the principles of the invention, there is disclosed a lighting system that allows for the mixing of light from a plurality of lighting sources, wherein the light is focused onto at a desired point or region.

(14) In accordance with the principles of the invention, there is disclosed a lighting system comprising a plurality of lighting sources arranged concentrically about a central or optical axis of a lens system wherein the lighting sources are oriented with respect to the central axis such that the

light emitted by the light sources targets the lens system at a point or region where the lens system projects the light to a desired focal point.

(15) In accordance with the principles of the invention, there is disclosed a lighting system comprising a plurality of lighting sources arranged substantially perpendicular to a central or optical axis of a lens system, wherein the light emitted by the lighting sources targets a light director that re-directs the emitted light to the lens system at a point or region where the lens system projects the light to a desired focal point.

(16) In accordance with the principles of the invention, there is disclosed a lighting device comprising a plurality of lighting sources arranged substantially skewed from an axis perpendicular to a central axis of a lens system, wherein the light emitted by the lighting sources targets a light director that re-directs the emitted light to the lens system at a point or region where the lens system projects the light to a desired focal point.

(17) In accordance with the principles of the invention, there is disclosed a lighting system comprising a plurality of lighting sources that are arranged to emit light individually, concurrently or sequentially in one of more light wavelength bands toward a light director, which redirects the light to a lens system that projects the light to a desired focal point.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) For a better understanding of exemplary embodiments and to show how the same may be carried into effect, reference is made to the accompanying drawings. It is stressed that the particulars shown are by way of example only and for purposes of illustrative discussion of the preferred embodiments of the present disclosure and are presented in the cause of providing what is believed to be the most useful and readily understood description of the principles and conceptual aspects of the invention. In this regard, no attempt is made to show structural details of the invention, or in a scale, in more detail than is necessary for a fundamental understanding of the invention, the description taken with the drawings making apparent to those skilled in the art how the several forms of the invention may be embodied in practice. In the accompanying drawings:

(2) FIG. 1A illustrates a front view of a first exemplary embodiment of a lighting device comprising multiple light sources in accordance with the principles of the invention.

(3) FIG. 1B illustrates a side view, through section A-A, of the first exemplary embodiment of the lighting device shown in FIG. 1A

(4) FIG. 2 illustrates a second side view, through section A-A, of the first exemplary embodiment of the lighting device shown in FIG. 1A.

(5) FIG. 3 illustrates a front view of a second aspect of the first exemplary embodiment shown in FIGS. 1A, 1B and 1C.

(6) FIG. 4A illustrates a front view of a first aspect of a second exemplary embodiment of a lighting device **400** in accordance with the principles of the invention.

(7) FIG. 4B illustrates a side view, through section B-B, of the first aspect of the second exemplary embodiment of a lighting device shown in FIG. 4A.

(8) FIG. 4C illustrates a front view of a second aspect of the second exemplary embodiment of a lighting device shown in FIG. 4A.

(9) FIG. 5A illustrates a side view of a first aspect of a third exemplary embodiment of a lighting device comprising multiple light sources in accordance with the principles of the invention.

(10) FIG. 5B illustrates a front view of a second aspect of the third exemplary embodiment shown in FIG. 5A.

(11) FIG. 6 illustrates a side view of a fourth exemplary embodiment of a lighting device comprising multiple light sources in accordance with the principles of the invention.

- (12) FIG. 7 illustrates a side view of a third aspect of the second exemplary embodiment of the lighting device shown in FIGS. 4A-4C.
- (13) FIG. 8 illustrates a side view of a fourth aspect of the second exemplary embodiment of the lighting device shown in FIGS. 4A-4C.
- (14) FIG. 9A illustrates a side view of a first aspect of a fifth exemplary embodiment of a lighting device in accordance with the principles of the invention.
- (15) FIG. 9B illustrates a front view of the first aspect of the fifth exemplary embodiment of the lighting device shown in FIG. 9A.
- (16) FIG. 9C illustrates a front view of a second aspect of the fifth exemplary embodiment of the lighting device shown in FIG. 9A.
- (17) FIG. 10 illustrates a front view of a third aspect of the fifth exemplary embodiment shown in FIG. 9A.
- (18) FIG. 11 illustrates a side view of a sixth exemplary embodiment of a lighting device in accordance with the principles of the invention.
- (19) FIG. 12 illustrates a front view of a head-mounted lighting configuration incorporating the lighting device disclosed herein.
- (20) FIGS. 13A-13C illustrate cut-away side views of examples of the lighting device shown in FIG. 8 in accordance with the principles of the invention.
- (21) FIG. 14 illustrates an expanded view of the area identified as FIG. 14 in FIG. 13A.
- (22) FIG. 15 illustrates a perspective view of a second aspect of a light director as shown in FIG. 8.
- (23) FIG. 16 illustrates a top view of the exemplary embodiment of the lighting device 815 shown in FIG. 15.
- (24) FIG. 17 illustrates a cut-away side view of a seventh exemplary embodiment of a lighting device comprising multiple light sources in accordance with the principles of the invention.
- (25) FIG. 18 illustrates a graph of an exemplary filtering of light emitted from lighting device 1800.
- (26) FIG. 19 illustrates a cut-away side view of a further aspect of the exemplary embodiment of a lighting device shown in FIGS. 8 and 15.
- (27) FIG. 20 illustrates a cut-away side view of a further aspect of the exemplary embodiment of a lighting device shown in FIG. 8.
- (28) FIGS. 21A, 21B illustrated exemplary aspects of the invention as discussed in FIG. 20.
- (29) It is to be understood that the figures and descriptions of the present invention described herein have been simplified to illustrate the elements that are relevant for a clear understanding of the present invention, while eliminating for purposes of clarity, many other elements. However, because these omitted elements are well-known in the art, and because they do not facilitate a better understanding of the present invention, a discussion of such elements is not provided herein. The disclosure, herein, is directed also to variations and modifications known to those skilled in the art.

DETAILED DESCRIPTION

- (30) FIG. 1A illustrates a front view of a first exemplary embodiment of a lighting device comprising multiple light sources in accordance with the principles of the invention.
- (31) In this illustrated front view, lighting device 100 comprises a housing 105 and a lens 102 through which lighting assembly 118 is visible. Lens 102 closes a distal end of lighting device 100.
- (32) Further illustrated are lighting sources 110 and 112 incorporated onto lighting assembly 118, wherein lighting sources 110, 112 are positioned concentrically around or about a central or optical axis 142, which is formed by lens 102.
- (33) Lighting sources 110 and 112 are oriented with respect to the central or optical axis 142 to direct the light output by the plurality of lighting modules to a point (not shown).
- (34) Lighting assembly 118 further includes control unit 116 that controls the operation of the lighting sources 110, 112 to emit light. Control unit 116 may comprise one or more of resistors,

transistors, diodes, capacitors that form dedicated hardware configuration and/or specialized hardware (e.g., ASIC, microcontroller, microprocessor) that enable control unit **116** to control the application of electrical energy to one or more of lighting sources **110**, **112**, etc.

(35) FIG. **1B** illustrates a side view, through section A-A, of the first exemplary embodiment of the lighting device shown in FIG. **1A**

(36) In this illustrated embodiment, lighting device **100** comprises lighting housing **105** into which lens assembly **101**, comprising at least one objective lens **102**, **104**, is positioned at a first end of lighting housing **105**. The at least one objective lens **102**, **104** are substantially concentric and define a central or optical axis **142** of lighting device **100**. The characteristics of lens **102**, **104** form a focal point **140** onto which light **103** passing through lens assembly **101** converges.

(37) Positioned at a second end of lighting housing **105** is lighting (or optical) assembly **118**, which comprises control unit **116** and lighting sources **110** and **112**. Connection between control unit **116** and lighting sources **110**, **112** is provided by arms or extensions **120**, **122**. Arms **120**, **122** provide a means for providing electrical energy to lighting sources **110**, **112** and further providing for adjustment of lighting sources **110**, **112** to change an angle of orientation of lighting source **110**, **112** with respect of central or optical axis **142**.

(38) As is further shown, lighting assembly **118** and lens housing **101** may be mountable within light device **100** by a screw-thread connection (i.e., **124/126** **150/151**, respectively). Although a screw-thread connection is shown, it would be understood that other connections methods may be employed without altering the scope of the invention. For example, a snap-fit or a bayonet connection may be utilized for lighting assembly **118** and/or lens assembly **101**.

(39) In accordance with one aspect of the invention, lighting sources **110**, **112** comprise holder **130**, electronic section (i.e., printed circuit board) **132**, light source **134** and lens **136**. Holder **130** provides a means for retaining lighting sources **110** (**112**) within lighting assembly **118**. Electronic section **132** may comprise a printed circuit board, including known components, such as resistors, transistors, capacitors, special hardware circuitry (e.g., ASIC), and/or microcontroller/microprocessor, which control a flow of electrical energy (voltage/current) to light element **134**. Light element **134** may comprise light emitting diodes (LEDs) that may possess lasing capability (i.e., semiconductor laser) or non-lasing capability (e.g., super-luminance diodes, etc.). Lens **136** provides for a concentration of the light emitted by light source **134**.

(40) A more detailed understanding of a preferred construction of lighting sources **110**, **112** may be found in the referred to related US patents wherein lighting sources **110** comprises at least one of: a lighting element (e.g., a light emitting diode), an aperture holder, an aperture and a dome lens, wherein the aperture holder and aperture are configured to adapt or configure the light outputted by the lighting element and the dome lens is configured to concentrate the light outputted through the aperture.

(41) For example, in the case of the emission of a white light, lighting element **134** may comprise a phosphor layer and a blue die positioned on the phosphor layer wherein the light emitted by light element **134** is primarily a white light. Aperture holder (not shown) may include a passthrough which is sized to allow the blue die element of light element **134** to passthrough, while blocking light emitted by the phosphor layer. An aperture, positioned on or within the aperture holder, may include a passthrough that may be sized to further reduce (adapt or configure) the amount of stray light of the phosphor layer to passthrough the aperture. In one aspect of the invention, the aperture passthrough may be sized to allow the blue die element of the lighting element **134** to passthrough. In another aspect of the invention, the aperture passthrough may be sized to allow only a center region of the blue die element of lighting element **134** to be viewable through the aperture passthrough and prevent the blue die element from being positioned within the aperture passthrough. In one aspect of the invention, the aperture holder passthrough may be substantially square or circular to allow the blue die element to passthrough, while the aperture passthrough may be circular or square and smaller than the blue die element. In addition, in accordance with the

principles of the teachings of the related US patents, the lighting element **134** may be within a focal length of the dome lens **136**.

(42) Although a specific configuration of lighting sources **110**, **112** is disclosed, it would be recognized that lighting source **110**, **112** may include only electronic section **132** and light element **134**, wherein at least one of the aperture holder, the aperture and dome lens **136** may not be utilized.

(43) Returning to FIG. **1B**, arm **120** provides electrical and mechanical connection from control unit **116** to lighting source **110**. For example, control unit **116** may provide electrical energy to lighting source **110** such that a light may be emitted by lighting source **110**. In one aspect of the invention, a magnitude (i.e., a drive current) of the electrical energy provided to lighting source **110** may be adjusted such that the light output may transition from a low output light to a high output light or from a light ON condition to a light OFF condition.

(44) Extension arm **120** may be adjustable in a manner that changes an orientation of lighting source **110** with respect of optical axis **142**.

(45) In this illustrated embodiment lighting source **110** is oriented with respect to an axis **182** wherein axis **182** is offset from an axis (not shown) that is substantially perpendicular to optical axis **142**.

(46) In one aspect of the invention, arm **120** may comprise a screw thread (not shown), which when turned in a first (e.g., clockwise) direction may decrease the angle of axis **180** with respect to optical axis **142**. Whereas, when turned in a second (e.g., counterclockwise direction), the angle of axis **180** with respect to optical axis **142** may increase.

(47) Alternatively, offset axis **180** may be preset in a manner to satisfy the angle characteristics presented herein, wherein a light emitted by lighting source **110** converges to a point a known distance from lens **102**.

(48) Further illustrated is second lighting source **112** positioned opposite optical axis **142**. Second lighting source **112** is similar in construction and electrical and physical operation to first lighting source **110** and a detailed discussion of the construction and operation of second lighting source **112** is not believed necessary, as those skilled in the art would understand both the construction and operation of second lighting source **112** from the discussion associated with lighting source **110**.

(49) Control unit **116** may further provide electrical energy to one or more of lighting sources **110**, **112** in a manner such that the light emitted by lighting sources **110**, **112** may be emitted concurrently, individually or sequentially.

(50) Although controller **116** is shown as an individual element, it would be understood that the control function of controller **116** may be incorporated into the electronic components associated with PCB **132**.

(51) Hence, the illustrated element **116** may merely comprise a base onto which arms **120**, **122** are positioned, and the control function may be included in PCB **132**.

(52) Accordingly, the lighting sources **110** and **112** may be referred to as lighting module **110/116** and **112/116** to illustrate the distributive nature of the electronic components associated with controller **116** and PCB **132**.

(53) FIG. **2** illustrates a second side view, through section A-A, of the first exemplary embodiment of the lighting device shown in FIG. **1A**.

(54) In this illustrated aspect of the invention of lighting device **100**, first lighting module **110/116** generates light directed along light path **220** toward lens assembly **101**. Similarly, second lighting module **112/116** generates light directed toward lens assembly **101** along light path **222**.

(55) The light generated by first lighting module **110/116** and second lighting module **112/116** passes through lens **104** and **102** and converges onto a known point **250**. In this case, known point **250** is illustrated to be approximately 16 inches from a lighting device **100**.

(56) Although sixteen inches is illustrated as the point of convergence **250** of the light emitted by lighting device **100**, it would be understood that the use of the measurement of 16 inches is only to

illustrate the convergence of the light and other distances or measurements from lighting device **100** may be achieved by changing the relationship between lighting modules **110/116**, **112/116** and lens housing **101**. Hence, other distances (or focal points) are considered within the scope of the invention claimed.

(57) FIG. 3 illustrates a front view of a second aspect of the first exemplary embodiment shown in FIGS. 1A, 1B and 1C.

(58) In this illustrated front view, a plurality of lighting modules (e.g., first lighting module **110/116**, second lighting module **112/116**, third lighting module **113/116** and fourth lighting module **114/116**) are positioned concentrically around optical axis **142**, wherein each of the plurality of lighting modules is attached to corresponding extensions (not shown), which are attached to base **116'**. Electrically energy (voltage, current) may be applied to the lighting modules **110/116** . . . **114/116**, through extensions (not shown) as previously discussed, wherein the electronic elements of controller **116/PCB 132** control the application of the voltage to corresponding lighting modules.

(59) The plurality of lighting modules **110/116** . . . **114/116** are oriented, as discussed, to direct the light outputted by the plurality of lighting modules to a focus point **250** (not shown).

(60) Although four (4) lighting modules are illustrated, it would be recognized that the number of lighting modules may be increased (i.e., greater than the illustrated 4) or decreased (i.e., 2 or 3) without altering the scope of the invention.

(61) FIG. 4A illustrates a front view of a first aspect of a second exemplary embodiment of a lighting device **400** in accordance with the principles of the invention.

(62) In this illustrated front view of lighting device **400**, a plurality of lighting sources and associated distributed controllers (referred to as first lighting module **110/116**, second lighting module **112/116**, third lighting module **113/116** and fourth lighting module **114/116**) are shown positioned along an inner circumference surface of lighting assembly **418**. Further illustrated is light director **415** extending from base **406** of lighting assembly **418**, wherein light director **415** is in a shape of a 4-sided pyramid. In addition, each of the sides or lateral faces of the illustrated 4-sided pyramid **415** may include a reflective surface (e.g., aluminum, mirror, etc.) **440**, **442**, **444**, **446**, that redirects light emitted by one or more of the lighting modules toward lens assembly **101**.

(63) In accordance with the principles of the invention, light emitted by lighting modules **110/116**, **112/116**, **113/116**, **114/116** is directed to a corresponding one of the lateral faces (which include reflective surfaces) such that the emitted light, which is in the plane of the paper, is redirected such that the emitted light is directed perpendicular to the plane of the paper. In accordance with the principles of the invention the light emitted by lighting modules **110/116** . . . **114/116** may be emitted individually, concurrently or sequentially.

(64) FIG. 4B illustrates a side view, through section B-B, of the first aspect of the second exemplary embodiment of a lighting device shown in FIG. 4A.

(65) In this second exemplary embodiment, lighting device **400** comprises a lighting housing **405** comprising lens assembly **101** positioned on a first end of lighting housing **405** and lighting assembly **418** positioned on a second, opposing end of light housing **405**. Lens assembly **101** comprises at least one objective lens **102**, **104** forming an optical axis **142** on which focal point **250** is formed, as previously discussed.

(66) Further illustrated is lighting assembly **418** comprising a first lighting module **110/116** (as first lighting source **110** and associated controller **116**) and second lighting module (second lighting source **112** and associated controllers **116**) positioned around or about an inner circumference to lighting assembly **418**. The illustrated first lighting source **110** and second lighting source **112** are similar in construction to lighting source **110**, **112** shown in, and described, with regard to FIGS. 1A, 1B and 2. Although only two lighting modules are shown, it would be recognized that a greater number of lighting modules may be distributed about the inner surface of lighting assembly **418**.

(67) Furthermore, although lighting assembly **418** is shown without a screw thread attachment to housing **405**, it would be understood that lighting assembly **418** may be removably attached to

lighting housing **405** through at least one of: a screw thread, a snap-fit, a bayonet, etc., type connection in a manner similar to that discussed with regard to lighting assembly **118** shown in FIG. **1B**, for example. Alternatively, lighting assembly **418** may be integrated into lighting housing **405**.

(68) In this first aspect of the second exemplary embodiment of a multi-light source lighting device, first lighting module **110/116** and second lighting module **112/116**, positioned along an internal circumference of lighting assembly **418**, emit or generate a light that is substantially perpendicular to optical axis **142** along light path **460**, **462**, respectively.

(69) Light director **415** operates to receive the emitted light and redirect the light emitted by first lighting module **110/116** and second lighting module **112/116** toward lens assembly **101**, along light paths **420**, **422**, respectively.

(70) In this illustrated embodiment, light director **415** comprises a pyramid shaped element positioned on base **406**, wherein the sides or lateral faces of light director **415** extend from base **406** at an angle oriented at substantially a forty-five (45) degree angle with respect to optical axis **142**; forming a ninety (90) degree angle at an apex of light director **415**, to allow light generated or emitted by first lighting module **110/116** and second lighting module **112/116** to be redirected toward lens assembly.

(71) Light director **415** further comprises reflective surfaces **440**, **442**, on at least a portion of corresponding one of its lateral surfaces, wherein reflective surfaces **440**, **442** operate to redirect a substantial portion of the light generated or emitted by first lighting module **110/116** and second lighting module **112/116** toward lens assembly **101**.

(72) In accordance with the principles of the invention, light directed along light paths **420** and **422** is outputted by lens assembly **101** such that the light converges onto known point **250**, as previously discussed.

(73) Although, lighting modules **110/116** and **112/116** are shown in a distributed controller configuration, it would be recognized that the controller **116** may be remote from lighting sources **110**, **112** as shown in FIG. **1B**, for example, without altering the scope of the invention claimed.

(74) FIG. **4C** illustrates a front view of a second aspect of the second exemplary embodiment of a lighting device shown in FIG. **4A**.

(75) In this illustrated front view, a plurality of lighting modules (e.g., first lighting module **110/116**, second lighting module **112/116**, and third lighting module **113/116**) are shown positioned along an inner circumference surface of lighting assembly **418**. Further illustrated is light director **415** extending from base **406** of lighting assembly **418**, wherein light director **415** is in a shape of a tetrahedron (i.e., a 3-sided pyramid). In addition, each of the lateral surfaces (or portions thereof) of light director **415** may include a reflective surface **441**, **442**, **443**, respectively, to reflect a substantial portion of the light emitted by corresponding ones of the lighting modules.

(76) As previously discussed, the light emitted by the lighting modules **110/116**, **112/116**, **113/116**, may be emitted concurrently, individually or sequentially.

(77) Although light director **415** shown in FIG. **4A** is shown as a 4-sided pyramid and shown in FIG. **4C** as a 3-sided pyramid, it would be recognized that a number of sides or lateral faces of light director **415** may be increased without altering the scope of the invention. For example, the number of lateral faces of the light director **415** may be determined based on a number of lighting modules positioned along the inner circumference surface of lighting assembly **418**. Alternatively, light director **415** may include an infinite number of sides (i.e., a cone or conical shape), which is independent of the number of lighting modules.

(78) FIG. **5A** illustrates a side view of a first aspect of a third exemplary embodiment of a lighting device comprising multiple light sources in accordance with the principles of the invention.

(79) In this third exemplary embodiment, light device **500** comprises a housing **505** and lens assembly **101** comprising at least one lens **102**, **104** forming an optical axis **142** onto which is formed focal point **250**.

(80) Further illustrated is lighting assembly **518** positioned at a second end of housing **506**. Lighting assembly **518** comprises first lighting module **510/116** (i.e., lighting source **510** and controller **116**) and second lighting module **512/116** (i.e., lighting source **512** and controller **116**) positioned along an inner circumference surface of lighting assembly **518** and a light director **515**. In this illustrated embodiment, first lighting source **510** and second lighting source **512** are similar to light sources **110**, **112** shown and described with regard to FIG. **1B**. However, lighting sources **510** and **512** lack one or more of the aperture holder, aperture, dome lens disclosed with regard to first lighting source **110** and second lighting source **112**. In this specific illustrated embodiment, lighting sources **510**, **512** lack dome lens **136**.

(81) In addition, lighting assembly **518** may, as with regard to lighting assembly **418**, be removably attachable to housing **505** or integrated into housing **505**.

(82) Light director **515** is similar to light director **415**, previously described. However, in this illustrated embodiment light director **515** is represented as a clipped pyramid or cone shaped element positioned on base **506**. That is, the term “clipped pyramid or clipped cone” represents a geometrical pyramid (or cone) in which a top portion has been removed. When viewed as a 3-dimensional object, light director **515** is in the form of a 3-dimensional trapezoidal shaped object.

(83) Light director **515**, which is oriented at a substantially 45-degree angle with respect to optical axis **142**, comprises reflective surfaces **440**, **442** that operate to redirect light generated by first lighting module **510/116** and second lighting module **512/116**, as discussed with regard to FIG. **4B**.

(84) Further illustrated is lens **570** positioned substantially perpendicular to optical axis **142**. In this illustrated example, lens **570** is positioned in contact with light director **515** and is sized such that light redirected from reflective surfaces **440**, **442** is captured by lens **570**.

(85) In accordance with this embodiment of the invention, light emitted by first lighting module **510/116** is directed along light path **560** and impinges upon reflective surface **440**, which redirects the light along light path **520** toward lens **570**. Similarly, light emitted by second lighting module **512/116** is directed along light path **562** and impinges upon reflective surface **442**. Reflective surface **442** redirects the light along light path **522** toward lens **570**.

(86) Lens **570** concentrates the light directed along light paths **520**, **522** toward lens assembly **101**, which converges the light onto known point **250**, as previously discussed.

(87) Although lens **570** shown in FIG. **5A** is depicted as extending a width of lighting assembly **518**, it would be understood that lens **570** may be included within a holder that extends the width of lighting assembly **518**, wherein the holder retains lens **570** in place and lens **570** may be sized to be sufficient to capture the light redirected by light director **515**.

(88) FIG. **5B** illustrates a front view of a second aspect of the third exemplary embodiment shown in FIG. **5A**.

(89) In this illustrated front view, a plurality of lighting modules (e.g., first lighting module **510/116**, second lighting module **512/116**, third lighting module **513/116** and fourth lighting module **514/116**) are shown positioned along an inner circumference surface of lighting assembly **518**. Further illustrated is light director **515** extending from base **506** of lighting assembly **518**, wherein light director **515** is in a shape of a 4-sided clipped pyramid.

(90) Further illustrated is lens **570** positioned substantially perpendicular to the optical axis **142** within holder **590**. Holder **590** extends the width of lighting assembly **518** and retains lens **570** in place.

(91) FIG. **6** illustrates a side view of a fourth exemplary embodiment of a lighting device comprising multiple light sources in accordance with the principles of the invention.

(92) In this fourth exemplary embodiment, lighting device **600** comprises housing **605** and lens assembly **101** comprising at least one lens **102**, **104** forming an optical axis **142**.

(93) Further illustrated is lighting assembly **618** comprising first lighting module **510/116** and second lighting module **512/116**, positioned along an inner circumference surface of lighting assembly **618** and light director **415**.

(94) First lighting module **510/116** and second lighting module **512/116** are similar to those elements described with regard to FIG. 5A and light director **415** is similar to the light director **415** described with regard to FIGS. 4A, 4B.

(95) Light director **415**, whose sides or lateral surfaces are oriented at a substantially 45-degree angle with respect to optical axis **142** comprises reflective surfaces **440**, **442**, which operate to redirect light emitted by first lighting module **510/116** and second lighting module **512/116**.

(96) Further illustrated is lens **570** positioned substantially perpendicular to optical axis **142**. Lens **570** is sized to capture light redirected from reflective surfaces **440**, **442** and direct the capture light toward lens assembly **101**.

(97) In this illustrated example, light emitted by first lighting module **510/116** directed along light path **660** and impinges upon reflective surface **440**, which redirects light toward lens **570** and lens assembly **101**, along light path **620**. Similarly, light emitted by second lighting module **512/116** directed along light path **562** impinges upon reflective surface **442**, which directs light toward lens **570** and lens assembly **101** along light path **622**.

(98) In accordance with the principles of the invention, light directed along light paths **620** and **622** is outputted by lens assembly **101** such that the outputted light converges onto known point **250**.

(99) Although lens **570** shown in FIG. 6 is depicted as extending the width of optical assembly **618**, it would be understood that lens **570** may be included within a holder that extends the width of optical assembly **618** in a manner similar to that shown in FIG. 5B.

(100) FIG. 7 illustrates a side view of a third aspect of the second exemplary embodiment of the lighting device shown in FIGS. 4A-4C, wherein the lighting sources **110**, **112** (or lighting modules **110/116**, **112/116**) are configured to cause light director **415** to redirect light to a known region **750** within lens assembly **101**.

(101) In this illustrated embodiment, lighting device **700** comprises lighting housing **405** comprising lens housing **101**, wherein lens housings **101** includes lens **102** and **104**, and lighting assembly **718** includes light director **415**, as previously discussed.

(102) In this illustrated embodiment, which is similar to the embodiments shown in FIGS. 4A, 4B, 4C, 5A, 5B, and 6, lens housing **101** may be removably attachable to lighting housing **405**, through the illustrated screw thread **715/716**. Although a screw thread connection is shown it would be recognized that lens housing **101** and lighting housing **405** may comprise other forms of connections as previously discussed. Similarly, lighting assembly **718** may be removably attachable to housing **405** as shown in FIGS. 1A, 1B, for example.

(103) Further illustrated are reflective surfaces **440** and **442**, which reflect light at an angle equal to the angle of the light incident to reflective surfaces **440**, **442**, with respect to an axis **740**, **742** normal to reflective surface **440**, **442**, respectively.

(104) In this third aspect of the embodiment shown in FIGS. 4A-4C, lighting sources **110** and **112** (or lighting modules **110/116**, **112/116**) are positioned along the interior surface of lighting housing **405** at an angle **710**, **712** with respect to optical axis **142**. Accordingly, the light emitted by lighting sources **110**, **112** (or lighting modules **110/116**, **112/116**) is not perpendicular to, but skewed (or slanted from) optical axis **142**.

(105) The angle at which light sources **110**, **112** (or lighting modules **110/116**, **112/116**) is offset from the optical axis **142** may be selected or determined to cause the light projected or emitted by lighting sources **110**, **112** (or lighting modules **110/116**, **112/116**) along light paths **460**, **462**, respectively, to be reflected from reflective surfaces **440**, **442** along paths **720**, **722** so as to be directed to known region **750** on lens assembly **101**.

(106) Region **750** may be determined, in part, based on the characteristics of lens **102**, **104** such that the light emitted by lighting sources **110**, **112** (or lighting modules **110/116**, **112/116**) converges at known point **250**, as discussed previously.

(107) FIG. 8 illustrates a side view of a fourth aspect of the second exemplary embodiment of the lighting device shown in in FIGS. 4A-4C.

(108) In this illustrated aspect of the second embodiment shown in FIGS. 4A-4C, lighting device **800** comprises lighting housing **405** comprising lens housing **101**, wherein lens housings **101** includes lens **102** and **104**, removably attachable to lighting housing **405**, and lighting assembly **818**, similar to the configuration shown in shown in FIGS. 4A, 4B, comprising lighting sources **110**, **112** (or lighting modules **110/116**, **112/116**) positioned about an inner circumference of light assembly **818**.

(109) Although, lighting assembly **818** is shown integrated into lighting housing **405**, it would be understood that lighting assembly **818** may similarly be removably attachable to lighting housing **405** as shown in FIGS. 1A, 1B, for example.

(110) In accordance with the principles of operation of this illustrated embodiment, light sources **110** and **112** (or lighting modules **110/116**, **112/116**) are arranged along the interior surface of lighting housing **405** (or lighting assembly **818**) such that light paths **460**, **462** are substantially perpendicular to optical axis **142**.

(111) Further illustrated is light director **815**, which is similar to light director **415**, including reflective surfaces **440**, **442**, which reflects light projected onto reflective surfaces **440**, **442**, at an angle equal to the angle of the light incident to light director **815**.

(112) In this illustrated embodiment, the peak angle **850** of light director **815**, as shown, is an obtuse angle (i.e., greater than 90 degrees) to enable the light emitted by lighting modules **110/116**, **112/116**, to be reflected toward region **750**.

(113) As is known in the art, the light reflected by a reflective surface is reflected at an angle, with respect to a normal (i.e., axis **740**, **742**) that is equal to the angle of incidence of light, wherein reflected light, in a manner similar to that discussed with regard to FIG. 7, is directed along light paths **720**, **722**.

(114) FIG. 9A illustrates a side view of a first aspect of a fifth exemplary embodiment of a lighting device in accordance with the principles of the invention.

(115) In this exemplary embodiment, lighting device **900**, which is similar to lighting device **400**, shown in FIGS. 4A, 4B, comprises a lighting housing **405** comprising a lens assembly **101** comprising at least one objective lens **102**, **104** and lighting assembly **918** comprising a plurality of lighting sources **110**, **112** (or lighting modules **110/116**, **112/116**) oriented about an inner circumference of lighting housing **405** and light director **815** comprising a peak angle **850** greater than ninety (90) degrees, as discussed with regard to FIG. 8, to redirect light contacting corresponding points on reflective surfaces **440**, **442** towards point **750** on lens assembly **101**. Furthermore, lens assembly **101** and lighting assembly **918** may be removably attachable to lighting housing **405**, as previously discussed.

(116) In accordance with the principles of this illustrated exemplary embodiment, lighting sources **110** and **112** (or lighting modules **110/116**, **112/116**) are positioned at different points along the inner circumference of lighting assembly **918**, such that light emitted by lighting source **110** (or lighting module **110/116**), projected along light path **460**, contacts light director **815** at a first point **961** and light emitted by lighting source **112** projected along light path **462** contacts light director **815** at a second point **962**.

(117) The light emitted by lighting source **110** (module **110/116**) and lighting source **112** module **112/116**) is re-directed toward region **750**, as previously discussed.

(118) In one aspect of the invention a plurality of lighting sources **110** (or lighting modules **110/116**) may be positioned about an inner circumference of lighting housing **405** in a first plane (in this illustrated case, the plane containing light path **460**) and a plurality of second lighting sources **112** (or modules **112/116**) may be positioned about an inner circumference of lighting housing **405** in a second plane (in this illustrated case, the plane containing light path **462**) wherein the first point of contact **961** of light emitted by lighting source **110** on reflective surfaces **440** is within the first plane, and the second point of contact **962** of light emitted by lighting modules **112/116** on reflective surfaces **442** is within the second plane.

(119) In one aspect of the invention, lighting sources **110** (modules **110/116**) may be configured to emit a white light (i.e., violet through red wavelength ranges) while lighting sources **112** (modules **112/116**) may be configured to emit light in one or more specific wavelength bands (e.g., ultra-violet, blue, green, yellow, orange, red, etc.).

(120) In another aspect of the invention, lighting sources **110** (module **110/116**) may comprise white light emitting elements, and selected ones of lighting sources **112** (module **112/116**) may comprise lighting elements emitting light in a wavelength range (e.g., an ultra-violet wavelength range) while selected other ones of lighting sources **112** (module **112/116**) may comprise lighting elements emitting light in a different wavelength range, e.g., blue wavelength range.

(121) FIG. **9B** illustrates a front view of the first aspect of the fifth exemplary embodiment of the lighting device shown in FIG. **9A**.

(122) In this illustrated aspect, a plurality of lighting sources **110** (or lighting modules **110/116**) are arranged about an inner circumference surface of lighting assembly **418** in a first plane. Similarly, a plurality of lighting sources **112** (or lighting modules **112/116**) are arranged about an inner circumference of lighting assembly **418**, as discussed with regard to FIG. **9A**.

(123) In this illustrated example, lighting modules **112/116** are not visible as these modules are positioned behind lighting modules **110/116** as the light emitted by lighting sources **110**, **112** are directed toward a same surface of light director **815**.

(124) As discussed, light emitted by lighting sources **110**, **112** (concurrently, individually or sequentially) is directed toward light director **815**, which redirects the light toward lens assembly **101** (not shown), as previously discussed.

(125) Although only four lighting sources **110** (or lighting module **110/116**) are shown, it would be recognized that the number of lighting modules included along the inner surface of lighting assembly **418** may be increased or decreased based on the number of reflective surfaces of light director **815**.

(126) FIG. **9C** illustrates a front view, of a second aspect of the fifth exemplary embodiment of the lighting device shown in FIG. **9A**.

(127) In this exemplary aspect, a plurality of lighting sources **110** and **112** (or lighting modules **110/116**, **112/116**) are arranged as pairs about an inner circumference surface of lighting assembly **418**, wherein the lighting source pair **110/112** are controlled by a same controller **116**.

(128) In this exemplary aspect, light emitted by lighting sources **110**, **112** (concurrently, individually or sequentially) is directed toward light director **815**, which redirects the light toward lens assembly **101** (not shown), as previously discussed.

(129) Although only four lighting modules **110/116** are shown, it would be recognized that the number of lighting modules included along the inner surface of lighting assembly **418** may be increased or decreased based on the number of reflective surfaces of light director **815**.

(130) FIG. **10** illustrates a front view of a third aspect of the fifth exemplary embodiment shown in FIG. **9A**.

(131) In this illustrated aspect of lighting device **900**, lighting device **900** comprises lighting housing **405** including lens housing **101** comprising lens **102** and lighting assembly **918** comprising light director **1015**, wherein light director **1015** is shown as a cone or conical element to re-direct light emitted by lighting sources **110**, **112** arranged along or about an inner circumference of lighting housing **405**. Conical element **1015** is advantageous as it represents a pyramid of an infinite number of sides.

(132) Lighting sources **110**, shown along (or about) an inner circumference of lighting housing **405** are arranged in a first plane and lighting sources **112** shown along (or about) an inner circumference of lighting housing **405** are arranged in a second plane (similar to the configuration shown in FIG. **9B**). Further illustrated are lighting source **112** offset from lighting modules **110/116**.

(133) In still a further aspect of the embodiment shown in FIG. **10**, both lighting sources **110** and

112 may be arranged on a same plane (i.e., the first plane and the second plane of FIG. 9B or a single plane of FIG. 4B) wherein the plurality of lighting modules **112/116** may be offset from the lighting modules **110/116**.

(134) In this further aspect of the embodiment shown in FIG. 9A, the use of conical shaped light direction **1015** allows for an increase in the number of lighting sources **110**, **112** that may be utilized to project light onto focal point **250**.

(135) FIG. 11 illustrates a side view of a sixth exemplary embodiment of a lighting device in accordance with the principles of the invention.

(136) In this illustrated embodiment, lighting device **1110** comprises lighting housing **405**, comprising lens housing **101** and a lighting assembly **1118**, similar to that shown in FIGS. 4A, 4B, wherein lens housings **101** includes lens **102** and **104** and lighting housing **1118** includes light director **415** and lighting sources **110** and **112** oriented at an angle **710**, **712** with regard to an inner surface of lighting assembly **1118**, as discussed with regard to FIG. 7. Furthermore, as discussed previously, one of lens housing **101** and lighting assembly **1118** may be removably attachable to housing **405** through one of a screw thread (**715/716**) a snap-fit connection, and a bayonet connection, etc.

(137) In accordance with the principles of operation of this illustrated embodiment, lighting assembly **1118** further includes a reverse conical or parabolic element **1110** including a passthrough **1115** and conical or parabolic section **1120** positioned at or near an apex of light director **415**. Reverse conical section **1110** includes reflective surface **1112** and conical section **1120** includes reflective surface **1122**.

(138) As discussed, the light emitted by lighting modules **110/116**, and **112/116**, in this case, is re-directed toward conic section **1120**. Conic section **1120** then reflects the received light back toward reverse conical section **1110**, which directs light towards lens assembly **101**.

(139) Lens assembly **101** receiving the light reflected off reflective surfaces **1112** converges the light toward known point **250**, as previously discussed.

(140) Although FIG. 11 represents a configuration similar to that disclosed with regard to FIG. 8, wherein the lighting modules **110/116**, **112/116** are shown substantially perpendicular to optical axis **142** from lighting assembly **1118**, it would be recognized that the principles of FIG. 11 may be applied to the configuration shown in FIG. 7, wherein lighting modules **110/116**, **112/116** are offset to optical axis **142**, without altering the scope of the invention claimed.

(141) FIG. 12 illustrates a front view of a head-mounted lighting configuration **1200** comprising an eyewear **1280** comprising a frame **1270** containing a left lens **1220a** and a right lens **1220b** into which are magnification devices **1225a**, **1225b**, respectively. Although the head mounted lighting device illustrated refers to an eyewear **1280**, it would be understood other types of head mounted light devices are considered within the scope of the invention. For example, U.S. Pat. No. RE 46,463, whose contents have been incorporated by reference, herein, illustrate head mountings, such as a head strap and headband, which are considered within the scope of the term "head mounted".

(142) Lighting configuration **1200** further comprises a battery assembly **1250**, an electronics section **1230** and a lighting device **100** as disclosed herein.

(143) Within lighting device **100** is shown, through lens **102**, lighting assembly **118** similar to that discussed with regard to FIGS. 1A and 1B.

(144) However, it would be understood by those skilled in the art that while the lighting device **100** shown in FIG. 12 is described with regard to the embodiment of the invention shown in FIGS. 1A, 1B, the lighting devices **400**, **500**, **600**, **700**, **800**, **900** and **1100** disclosed, herein, may be identified as lighting device **100** in relation to the configuration shown in FIG. 12. Hence, the illustrated lighting assembly **118** may similarly be referred to as lighting assemblies **418**, **518**, **618**, etc.

(145) In accordance with the principles of the invention, the light output from lighting device **100** may be selected to provide one or more different light outputs based on the composition of the

lighting sources incorporated into lighting modules **110/116**, **112/116**, etc.

(146) Further illustrated is a sensing unit **1285** positioned along a top of battery assembly **1250**.

Sensing unit **1285** may be one of a contact sensor, such as a capacitive touch sensor or a contactless sensor, such as an infra-red (IR) sensor, an ultra-sonic sensor, a proximity sensor, and other similar devices. Although not shown it would be appreciated that sensing unit **1285** may be positioned on or in electrical contact with electronics section **1230**.

(147) Battery assembly **1250** incorporates a battery, therein, (not shown) that provides power (electrical energy in the form of a voltage and/or current) to lighting device **100**. Although the battery is disclosed with regard to battery assembly **1250**, it would be understood that a battery or other source (e.g., AC/DC power converters) providing electrical energy to the lighting sources **110**, **112** (or **512**, **514**), etc., for example, may be remote from the head mounted lighting device **1200**. See, for example, U.S. Pat. No. RE46,463, whose contents have been incorporated by reference, herein.

(148) Electronic section **1230** includes circuitry (not shown) that controls the application of the electrical energy (i.e., voltage/current) from the battery (not shown) contained within the illustrated battery assembly **1250** to lighting assembly **118** in lighting device **100**. Information from sensing unit **1285** to the circuitry within electronic section **1230** may also provide information suitable for controlling the state of the lighting modules **110/116**, **112/116**, etc., within lighting element **100**.

(149) In accordance with the principles of the invention, light output by one or more of the lighting modules **110/116**, **112/116** (**510/116**, **512/116**) may be controlled by operation of the sensing element **1285**, for example. In one aspect of the invention, wherein each of the lighting modules **110/116**, **112/116** (**510**, **512**) generates a white light, sensing element **1285** may operate to turn ON or turn OFF the light generated by each of the modules **110/116**, **112/116** (**510**, **512**), individually, sequentially or concurrently. Similarly, in an exemplary configuration wherein first lighting module **110/116**, generates light in an ultra-violet (UV) wavelength range and second lighting module **112/116** generates light in a visible (e.g., a blue, a green, a yellow, an orange, a red or a white) wavelength range, sensing element **1285** may operate to turn ON or turn OFF selected ones of the first lighting module **110/116** and second lighting module **112/116** so as to generate different light outputs.

(150) FIGS. **13A-13C** illustrate cut-away side views of examples of the lighting device shown in FIG. **8** in accordance with the principles of the invention.

(151) In these illustrated exemplary embodiments of lighting configurations disclosed herein, light directed toward region **750** is, after passing through lens **102**, **104**, is focused at a point **250** a known distance from lighting sources **110**, **112**; the known distance being determined in part by the illustrated apex angle associated with light director **815** and optical characteristics of lens **102**, **104**.

(152) FIG. **13A** illustrates a lighting configuration similar to that shown in FIG. **8**, wherein the apex angle of light director **815** set at 90.44 degrees (i.e., greater than 90 degrees) such that the light emitted by corresponding ones of the lighting sources **110**, **112** (or lighting modules **110/116**, **112/116**) is re-directed toward region **750** on lens **104**, **102** along optical paths **720**, **722**, as previously discussed.

(153) In accordance with the principle of the invention, region **750** is selected to enable the light emitted by corresponding ones of the lighting sources **110**, **112** (or lighting modules **110/116**, **112/116**), and re-directed by light director **815**, to be focused on point **250** at a known distance from lens **102**, **104** and/or lighting sources **110**, **112** (or lighting modules **110/116**, **112/116**) after passing through lens **102**, **14**. In this illustrated example, light emitted by lighting sources **110**, **112** (or lighting modules **110/116**, **112/116**) is directed along light paths **720**, **722**, respectively, wherein the light along light paths **720**, **722** after passing through lens **102**, **104** is focused onto a point **250**, which is selected as 16 inches from the light sources **110**, **112**.

(154) FIG. **13B** illustrates a lighting configuration, similar to that shown in FIG. **8**, wherein the angle of light director **815** set at 90.27 degrees (i.e., greater than 90 degrees), such that light emitted

by lighting sources **110**, **112** is re-directed toward region **750** on lens **104**, **102** along optical paths **720**, **722**, as discussed with regard to FIG. **13A**.

(155) In accordance with the principle of the invention, region **750** is selected to enable the light emitted by corresponding ones of the lighting sources **112/116**, and re-directed by light director **815**, to be focused on point **250** a known distance from lens **102**, **104** and/or lighting sources **110**, **112**. In this illustrated example, light emitted by lighting sources **110**, **112** (or lighting modules **110/116**, **112/116**) is focused onto point **250**, which in this illustrated example is selected as 20 inches from the light sources **110**, **112**.

(156) FIG. **13C** illustrate a lighting configuration, similar to that shown in FIG. **8**, wherein the angle of light director **815** set at 90.18 degrees (i.e., greater than 90 degrees), wherein the illustrated angle is set to re-director light toward region **750** on lens **104**, **102** along optical paths **720**, **722**, as discussed with regard to FIG. **13A**.

(157) In accordance with the principle of the invention, region **750** is selected to enable the light emitted by corresponding ones of the lighting sources **110**, **112** (or lighting modules **110/116**, **112/116**), and re-directed by light director **815**, to be focused on point **250** at a known distance from lens **102**, **104** or lighting sources **110**, **112**. In this illustrated example, light emitted by the lighting sources **110**, **112** is focused onto point **250**, which in this illustrated example is selected to be 23 inches from the light sources **110**, **112**, in a manner similar to that discussed above.

(158) FIG. **14** illustrates an expanded view of the area identified as FIG. **14** in FIG. **13A**.

(159) As would be known in the art, light, the direction of light passing through a media of different reflective indices (e.g., air-glass; glass-air) is altered with respect to an axis that is normal (i.e., perpendicular) to the lens. The degree of alteration in the direction of the path being determined by at least one optical characteristic associated with at least one objective lens. More specifically, the degree of alteration of the direction of the path of light passing through the objective lens is based on at least a ratio of the indices of reflectivity of the different media. Hence, region **750** is selected such that the direction of light travelling along optical paths **720**, **722**, and falling within region **750**, is directed by illustrated lens **104** along light paths **1420**, **1422** to be focused onto an object or plane (not shown) at a desired distance from the lighting sources.

(160) In this illustrated accordance with the principles of the invention, lens **104**, for example, comprises a compound (e.g., bi-convex, aspheric, etc.) lens, wherein light passing along light paths **720**, **722** is refracted, by the difference in the reflective indices of the material, to exit lens **104** along light paths **1422**, **1424**. Accordingly, light paths **1422**, **1424** are shown being in a non-parallel relation with respect to optical axis **142** to allow for the convergence of light along light paths **1422**, **1424** onto a point **250** (i.e., an object or plane a known distance from lens **104**).

(161) A determination of the apex angle of light director **815** may consider at least one of: Index of refraction of the material of the lens; a radius (power) of the lenses that the light emitted by the lights sources passes through; a location of where the light passes through the lenses with respect to an optical axis of the lenses; an index of refraction of lenses: a distance of the lenses with respect to the light director; and a focal lengths of the desired distance to focus the light.

(162) FIG. **15** illustrates a perspective view of a another embodiment of a light director similar to that shown in FIG. **8**.

(163) In this illustrated aspect, light director **1515** comprises lighting assembly **1518**, which is similar to lighting assembly **818** shown in FIG. **8**, comprises a plurality of reflective surfaces **1540**, **1542**, which include at least one of a mirror surface, and a highly polished metal surface (e.g., aluminum), etc.) arranged in an angular relationship extending from base of lighting assembly **1518** at an angle. In one aspect of the invention, the plurality of reflective surfaces **1540**, **1542** may extend from the base at an acute angle such that an obtuse apex angle is formed. Further illustrated are a plurality of lighting sources **110**, **112** (or lighting modules **110/116**, **112/116**) arranged on an inner circumference of assembly **1518**, such that light projected by lighting sources **110**, **112**, is substantially perpendicular to optical axis **142**, similar to the configuration shown in FIG. **8**.

(164) In accordance with this aspect of the invention, light director **1515** comprises surfaces **1540**, **1542** (which are similar to reflective surfaces **440**, **442**) arranged such that angle (i.e., an apex angle) between surfaces **1540**, **1542** is greater than ninety (90) degrees (i.e., obtuse), as discussed with regard to FIG. **8** as and light director **815**.

(165) In this aspect of the invention, light emitted by lighting modules **110/116**, **112/116** (is re-directed by surfaces **1540**, **1542** toward lens assembly **101** (i.e., lenses **102**, **104**) and converged onto point **250**, a known distance from lens assembly **101** as discussed with regard to FIGS. **8**, and **13A-13C**. The selected angle of the highly reflective surfaces **1540**, **1542** may be fixed in place by an adhesive (e.g., epoxy) to retain reflective surfaces **1520**, **1524** in place.

(166) Although the light director **1515** is discussed as being comparable to the light director **815** and the lighting sources **110**, **112** configured shown in FIG. **8**, it would be recognized that configuration of light director **1515** (i.e., a plurality of highly reflective surfaces) would be applicable to the configuration shown in FIG. **7**, wherein the apex angle between surfaces **1540**, **1542** is a perpendicular angle and the lighting sources **110**, **112** are arranged at an angle with respect to the optical axis **142** (not shown).

(167) Furthermore, although surfaces **1540**, **1542** are illustrated as being planar, it would be within the knowledge of those skilled in the art to incorporated surfaces that are spherical or parabolic to direct light as discussed herein.

(168) Although the configuration shown in FIG. **15** is discussed with reference to FIG. **8**, it would be understood by those skilled in the art that the principles discussed with regard to FIG. **7** would be applicable to a configuration of a light director as discussed with regard to FIG. **15**.

(169) FIG. **16** illustrates a top view of the exemplary embodiment of the lighting assembly **1518** shown in FIG. **15**.

(170) In this illustrated view, reflective surfaces **1540**, **1542**, extending from a base of assembly **1518** are joined together to form an apex angle **850**.

(171) Further illustrated is an adjustment mechanism **1620** positioned between the reflective surfaces **1540**, **1542**. Adjustment mechanism **1620** may be configured to alter the orientation of reflective surfaces **1540**, **1542** with respect to each other, such that apex angle **850** between the reflective surfaces **1540**, **1542** may be altered.

(172) In one aspect of the invention, adjustment mechanism **1620** may comprise a screw thread that allows for the expansion or contraction of apex angle **850** formed by surfaces **1540**, **1542**. After a desired apex angle is achieved, reflective surfaces **440**, **442** may be retained in place by an epoxy.

(173) Accordingly, one skilled in the art would, with the adjustment device shown in FIG. **16**, may determine apex angle **850** with respect to focusing light onto point **250**, when one or more of the distance to point **250** is changed or when one or more of the characteristics of the lens **102**, **104** in lens assembly **101** is changed.

(174) Although, the adjustment device shown in FIG. **16** is presented as a screw thread device, it would also be known in the art that the housing containing the plurality of reflective elements may be prefabricated to position the reflective elements at a known angle when the optical characteristics of the at least one lens is known and fixed.

(175) FIG. **17** illustrates a cut-away side view of a seventh exemplary embodiment of a lighting device comprising multiple lights sources in accordance with the principles of the invention.

(176) In this illustrated embodiment of light device **1700**, which is similar to that discussed with regard to FIGS. **7** and **8**, comprises lens housing **101** including lens **102**, **104** and lighting housing **405**, in which comprises lighting assembly **1718**. Lighting assembly **718** comprises a plurality of lighting sources **1710**, **1712** (or lighting modules **1710/116**, **1720/116**, which are comparable to lighting sources **110**, **112**, previously discussed. Further illustrated is light director **1715**, which as discussed with regard to light director **415** is orientated at a 45 degree angle with respect of base **405**.

(177) In accordance the principles of this exemplary embodiment of the invention, lighting sources

1712 (or lighting module **1712/116**) is positioned along an inner circumference of assembly **1718** and lighting source **1710** (or lighting module **1710/116**) is positioned on base element **405** of light assembly **1718**, substantially along optical axis **142**. Accordingly, light emitted by lighting source **1710** is projected along optical axis **142** toward point **250**. In one aspect of the invention, lighting source **1710**, similar to lighting source **110**, may emit a white light, which lighting source **1712**, similar to lighting source **112**, may emit a non-white light (e.g., Ultra-violet, blue, red, infra-red, etc.)

(178) Light director **1715**, similar to light director **415**, comprises reflective surface **1716** configured to receive light **1762** emitted by lighting source **1712**, and re-direct the emitted light along optical axis **142**, wherein light emitted by lighting sources **1710** and **1712** are combined into a single light projected along optical axis **142**.

(179) in addition, reflective surface **1716** may include an optical coating **1717**, which allows for the transmission of light **1760** emitted by lighting source **1710** (or lighting module **1710/116**), such that light from lighting sources **1710**, while reflecting light emitted by lighting source **1712**. In this manner light emitted by lighting sources **1710** and **1712** are projected onto common point **250** positioned along the optical axis **142**.

(180) Accordingly, light director **1715** operates as a beam combiner to allow the transmission of 100 percent of the light emitted by lighting source **1710** and 100 percent of the light emitted by lighting source **1712** to be directed along optical axis **142**.

(181) In one aspect of the invention, optical coating **1717** may further include a filter characteristic that may be utilized to structure or limit a wavelength range of the light emitted by lighting device **1700**. In an alternative configuration, lighting device **1700** may include filter **1742** positioned on a distal end of lighting device **1700**. Filter **1742**, similar to the filter characteristics of optical coating **1717** may be utilized to structure or limit a wavelength range of the light emitted by lighting device **1700**.

(182) Although FIG. **17** is discussed with regard to a configuration wherein a white light source is positioned on base **406** and a non-white source is positioned on an inner circumference of housing **405**, it would be within the knowledge of those skilled in the art to alter the positioning of the lighting sources to achieve a combined light output based on the teachings of the invention presented herein.

(183) FIG. **18** illustrates a graph of an exemplary filtering of light emitted from lighting element **1700**.

(184) In this illustrated example, optical coating **1717** (or filter **1742**) shown in FIG. **17**, is configured to allow the passage of light emitted by device **1700** to a first wavelength range, wherein the first wavelength is in range of 400 and 670 nm. Hence, while lighting source **1710** may emit a white light (i.e., 400-700 nm), the light emitted by lighting device **1700** is limited to a range of 400-60 nm. In addition, optical coating **1717** (or filter **1742**) may be further configured to allow passage of light within a range of 730-780 nm. Hence, while lighting source **1712** may emit an infra-red light (wavelengths greater than 700 nm), filter **1742** limits the light emitted by lighting device **1700** to be within a range of 730-780 nm.

(185) Thus, the light projected onto point **250** is limited to wavelength ranges that are allowed to pass through coating **1717** and/or filter **1742**.

(186) In one aspect of the invention, a combination of coating **1717** and filter **1742** may be utilized to allow passage or emission of a light having desired wavelength ranges for the combination of light.

(187) In this illustrated example, the wavelength ranges associated with coating **1717** (or filter **172**) are arbitrarily chosen merely to describe the optical characteristics of the coating or filtering. It would be within the knowledge of those skilled in the art to select other band-pass wavelength configurations without undue experimentation to achieve a light emitted by the devices discussed, herein, to have other desired wavelength characteristics.

(188) Although, the use of optical coating and filters is disclosed with regard to lighting device **1700**, it would be recognized that the principles of the illustrated optical coating and/or filtering may be applied to each of the lighting devices disclosed, herein.

(189) FIG. **19** illustrates a cut-away side view of a further aspect of the exemplary embodiment of a lighting device shown in FIGS. **8** and **15**.

(190) In accordance with this aspect of the invention, which is comparable to the embodiment disclosed in FIGS. **8** and **15**, lighting device **1900** comprises lens assembly **101** and lighting assembly **1518**, comprising light director **1515** and a plurality of lighting sources **110**, **112** (**1710**, **1712**) (or lighting modules **110/116**, **112/116**). In accordance with this aspect of the invention, light director **1515** comprises a plurality of reflective surface **1540**, **1542** arranged at an apex angle **850** wherein in one aspect of the invention, the apex angle is greater than 90 degrees.

(191) An understanding and more detailed description of the aspect of this invention may be obtained from a reading of the disclosure with regard to FIGS. **18** and **15**.

(192) In addition, in this aspect of the invention, reflective surfaces **1540**, **1542** fail to meet at their apex. Rather, surfaces **1540**, **1542** are sized to create a gap **1930** at their apex through which light may pass.

(193) Further illustrated is lighting sources **110**, **112** (**1710**, **1712**) (or lighting modules **110/116**, **112/116**), positioned on an inner circumference of housing **405**. Lighting sources **110**, **112** (modules **110/116**, **112/116**) as previously discussed with regard to FIGS. **8** and **15**, emit light that is reflected by reflecting surfaces **1540**, **1542**, respectively, wherein the reflected light is projected along light path **720**, **722**, respectively, onto region **750**. As discussed more fully with regard to FIG. **14**, for example, light within region **750**, after passing through lens assembly **101** is focused onto point **250**, at a known distance from lens assembly **101**.

(194) Further illustrated is an additional lighting source **1710** (module **1710/116**) positioned on base **405**. As previously discussed with regard to FIG. **17**, is configured to emit a light along optical axis **142** formed by lens **102**, **104**. In this illustrated example, light emitted by lighting source **1710** passes unaltered through gap **1930** to project light onto region **750**.

(195) In accordance with the principles of the invention, light emitted by lighting sources **110**, **112**, and **1710**, may be combined within region **750**, wherein the combined light is focused onto point (i.e., object, plane) **250**, as previously disclosed.

(196) In one aspect of the invention, lighting source **1710** may emit a white light and lighting sources **110**, **112** may emit light in one or more colored light wavelength bands (i.e., non-white). For example, although lighting source **110** has been described with regard to emitting a white light, in the previously discussed devices, it would be understood that the disclosed light emitted by lighting sources **110**, **112**, **114**, **1710**, **1712** may be altered without altering the scope of the invention claimed.

(197) For example, lighting source **1710** may emit a white light and lightings sources **110**, **112** may emit a same colored light (e.g., blue light, green light, red light, etc.), or may emit light within a same color wavelength band (i.e., blue light wavelength band, green light wavelength band, etc.) or within different colored light wavelength bands. Or lighting source may emit light in a colored wavelength band and lighting sources **110**, **112** may emit light in a same or different colored wavelength band.

(198) In one aspect of the invention, lighting element **1900** may further comprise one or more filter elements **1940** (**1942**) that may be used to limit one or more wavelength ranges emitted by lighting element **1900** similar to that discussed with regard to FIGS. **17** and **18**, for example. In one aspect of the invention filter **1940** may be inserted within, or proximate to, gap **1930** between reflective surfaces **1542**, **1544**. In this configuration, light emitted by lighting source **1710** may be limited in a desired wavelength range. For example, blue light wavelength range or limited white light wavelength range similar to that shown in FIG. **18**.

(199) Alternatively, filter **1942** may be inserted in a distal end of lens assembly **101**. In this

alternative configuration, light emitted by the combination of lighting sources **1710**, **110**, **112**, etc., may be limited to one or more known wavelength ranges (e.g., see FIG. **18**).

(200) FIG. **20** illustrates a cut-away side view of a further aspect of the exemplary embodiment of a lighting device shown in FIG. **8**.

(201) In this exemplary aspect of the invention, lighting device **2000** including lens assembly **101** and housing **405**, which includes lighting assembly **2018** comprising lighting sources **110**, **112** (modules **110/116**, **112/116**) positioned on an inner surface associated with lighting assembly **2018** and lighting module **1710/116** positioned on base **406**, similar to the configuration shown in FIG. **19**.

(202) Further illustrated is light director **2015**, which similar to light director **815**, previously discussed, comprises reflective surfaced **440**, **442** orientated at an angle with respect to base **405**, wherein reflective surfaces **440**, **442** merge at an apex; the angle at which similar to light director **815** is greater than ninety (90) degrees. In this this aspect of the invention, light director **2015** further comprises passage or tunnel **2030**, extending from the illustrated apex to base **406**, wherein light emitted by lighting module **1710/116** passes to lens **102**, **104**.

(203) In one aspect of this configuration, light director **2015** operates in a manner similar to that discussed with regard to light director **815** and **1515** (see FIGS. **8**, **15** and **19**) in that light from lighting sources **110**, **112** is reflected by reflective surfaces **442**, **444** toward region **750** and light emitted by lighting source **1710** is directed to region **750** on lens **104**, **102** along optical axis **142**. The combined light may converge onto point **250** in a manner as previously discussed.

(204) Further illustrated are optional, filters **1940**, **1942** that may limit a wavelength range of the light emitted by lighting device **2000** as discussed previously, with regard to FIGS. **17**, **18** and **19**.

(205) FIGS. **21A**, **21B** illustrated exemplary aspects of the invention discussed in FIG. **20**.

(206) FIG. **21A** illustrates a front view of the first aspect of the lighting device shown in FIG. **20**.

(207) In this illustrated aspect, a plurality of lighting sources **110**, **112** (or lighting modules **110/116**, **112/116**) are arranged about an inner circumference surface of lighting assembly **2018** in a first plane.

(208) Light emitted by lighting sources **110**, **112** (concurrently, individually or sequentially) is directed toward light director **2015**, which redirects the light toward lens assembly **101** (not shown), as previously discussed.

(209) Further illustrated is lighting source **1710** (or lighting module **1710/116**) viewable through passage **2030** within light director **2023**. Lighting source **1710** emits light along optical axis **142** (not shown as it extends into the plane of the page the drawing is on) toward lens assembly **101** as previously discussed. Although only four lighting sources **110**, **112** (or lighting module **110/116**, **112/116**) are shown, it would be recognized that the number of lighting modules included along the inner surface of lighting assembly **2018** may be increased or decreased based on the number of reflective surfaces of light director **2015**. In addition, although a single lighting source **1710** is discussed, it would be understood that lighting source **1710** (and lighting sources **110**, **112**, **114**) may comprise an array of lighting sources that may emit light in one of: a same or a different wavelength band.

(210) FIG. **21B** illustrates a front view, of a second aspect of the lighting device shown in FIG. **20**.

(211) In this exemplary aspect, a plurality of lighting sources **110** and **112** (or lighting modules **110/116**, **112/116**) arranged as pairs about an inner circumference surface of lighting assembly **2018**, wherein the lighting source pair **110/112** are controlled by a same controller **116**.

(212) In this exemplary aspect, light emitted by lighting sources **110**, **112** (concurrently, individually or sequentially) is directed toward light director **2015**, which redirects the light toward lens assembly **101** (not shown), as previously discussed.

(213) Although only four lighting modules are shown, it would be recognized that the number of lighting modules included along the inner surface of lighting assembly **2018** may be increased or decreased based on the number of reflective surfaces of light director **815**.

(214) Although the lighting devices discussed have been described with regard to two lighting modules, it would be understood that the multi-source lighting configurations shown, herein, may include a plurality of lighting sources suitable for emitting light of different light wavelength ranges (e.g. UV, violet, blue, green, yellow, orange, red, IR, etc.).

(215) In summary, a lighting device is disclosed, which comprises a plurality of lighting modules or sources arranged at an offset angle from an optical axis of a lens assembly, wherein the light is directed toward a lens assembly in a manner that allows for the convergence of the light from the plurality of lighting sources upon a known point. In a second embodiment, a lighting device is disclosed comprising a plurality of lighting modules or sources arranged along an inner circumference surface of the device house and the light generated by the lighting sources is directed to a device for redirecting the light toward a lens assembly in a manner that allows for the convergence of the light from the plurality of lighting sources to a desired point.

(216) In accordance with one aspect of the principles of the invention, the light director, receiving light from the lighting modules or sources, may be configured to re-direct the light to a known region to enable the re-directed light to converge to a known point after passing through at least one projection (or objective) lens. In accordance with another aspect of the invention, the lighting sources may be oriented with respect to the light director such that the light emitted by the lighting sources are re-directed by the light director such that the re-directed light is focused onto a known point on the objective lens.

(217) Although the invention has been described with regard to a lighting source, in a preferred embodiment the lighting sources are light emitting diodes (i.e., LEDs), it would be understood that other lighting sources may be incorporated into the invention disclosed without undue modification and, thus, considered within the scope of the invention claimed.

(218) In addition, while the term “lighting module” or “lighting source” has been described with regard to “light emitting diode,” it would be recognized that the term “a light emitting diode” may comprise a plurality of light emitting diode elements arranged in one of a matrix or circular pattern. Furthermore, the plurality of light emitting diodes may be within a base dimension of the disclosed dome lens and the positioning of the at least one light emitting diode within a focal length of the dome lens is advantageous as it de-focuses and blurs the distinction of the light emitted by the individual light emitting diodes within the array or circular pattern.

(219) Furthermore, the light emitted or output by the light emitting diode(s) is referred to being within a frequency or wavelength range, wherein the wavelength range represents a color (e.g., blue light approximately 400 nanometers to 450 nanometers). However, it would be further understood that light, particularly with regard to light emitting diodes, may be further measured in color temperature, as degrees Kelvin. For example, a soft white light may be measured in a range of 2700° K-3000° K whereas a bright white light may be expressed in a range of 5000° K-6000° K.

(220) Hence, one of ordinary skill in the art would appreciate that the reference to light or emitted light may be measured as frequency, wavelength, color or color temperature, and such terminology of light is incorporated and used interchangeably, herein.

(221) One of ordinary skill in the art will further recognize and appreciate that various modifications and changes can be made without departing from the scope of the invention as set forth in the claims. Accordingly, the specification is to be regarded in an illustrative manner, rather than with a restrictive view, and all such modifications are intended to be included within the scope of the invention. Benefits, other advantages, and solutions to problems have been described above with regard to specific embodiments. The benefits, advantages, and solutions to problems, and any element(s) that may cause any benefits, advantages, or solutions to occur or become more pronounced, are not to be construed as a critical, required, or an essential feature or element of any or all of the claims.

(222) As used herein, the terms “comprises”, “comprising”, “includes”, “including”, “has”, “having”, or any other variation thereof, are intended to cover non-exclusive inclusions. For

example, a process, method, article or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. In addition, unless expressly stated to the contrary, the term “or” refers to an inclusive “or” and not to an exclusive “or”. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present); A is false (or not present) and B is true (or present); and both A and B are true (or present).

(223) The terms “a” or “an” as used herein are to describe elements and components of the invention. This is done for convenience to the reader and to provide a general sense of the invention. The use of these terms in the description herein should be read and understood to include one or at least one. In addition, the singular also includes the plural unless indicated to the contrary. For example, reference to a composition containing “a compound” includes one or more compounds. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.

(224) All numeric values are herein assumed to be modified by the term “about,” whether or not explicitly indicated. The term “about” generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (i.e., having the same function or result). In any instance, the terms “about” may include numbers that are rounded (or lowered) to the nearest significant figure. Furthermore, the values presented herein are merely to illustrate the concepts and are not to be considered as the only values that have been contemplated and considered.

(225) It is expressly intended that all combinations of those elements that perform substantially the same function in substantially the same way to achieve the same results are within the scope of the invention. Substitutions of elements from one described embodiment to another are also fully intended and contemplated.

Claims

1. A lighting device comprising: a lens assembly comprising: at least one objective lens, each of the at least one objective lens including at least one optical characteristic, wherein the at least one objective lens is configured to form an optical axis passing through an axis of the at least one objective lens; and a lighting assembly comprising: a plurality of lighting sources positioned about an inner circumference of the lighting assembly, each of the plurality of lighting sources configured to: emit light along an axis toward the optical axis; at least one lighting source positioned on a base of the lighting source, the at least one lighting source configured to: emit light along the optical axis; and a plurality of reflective surfaces arranged opposite the plurality of lighting sources arranged about the inner circumference and extending, at an angle, from the base of the lighting assembly toward an apex, wherein an angular relation of the multiple reflective surfaces forming an apex angle therebetween, the plurality of reflective surfaces are configured to: receive the light emitted by at corresponding ones of the plurality of lighting sources positioned about the inner surface of the lighting assembly; re-directed the received light toward a known region associated with the at least one objective lens, and pass the light emitted by the at least one lighting source positioned on the base through an opening formed at the apex of the plurality of reflective surfaces toward the known region associated with the at least one objective lens, wherein the optical characteristics associated with each the at least one objective lens causing the light within the known region to be focused to a point a known distance from the at least one objective lens.
2. The lighting device of claim 1, wherein the plurality of lighting sources comprising: a light emitting diode; and at least one of: an aperture, an aperture holder and a dome lens.
3. The lighting device of claim 1, comprising: a controller configured to: control an emission of light from the plurality of lighting sources.
4. The lighting device of claim 1, comprising: a plurality of controllers, wherein each of the plurality of controllers configured to: control an emission of light from a corresponding one of the

plurality of lighting sources.

5. The lighting device of claim 1, wherein the plurality of reflective surfaces comprise one of: a plurality of mirrors and a plurality of highly polished surfaces.

6. The lighting device of claim 1, wherein the apex angle is greater than ninety (90) degrees and light emitted by the plurality of lighting sources is emitted along an axis substantially perpendicular to the optical axis.

7. The lighting device of claim 1, wherein the apex angle is substantially equal to ninety (90) degrees and light emitted by the plurality of lighting sources is emitted at an angle offset from an axis substantially perpendicular to the optical axis.

8. The lighting device of claim 1, wherein the plurality of reflective surfaces form a prism positioned on the base of the light assembly.

9. The lighting device of claim 8, wherein the prism comprises: a passthrough extending from the gap at the apex of the plurality of reflective surfaces through the prism to the base.

10. The lighting device of claim 9, comprising: a filter positioned within the gap, the filter configured to allow light within at least one wavelength range to pass through.

11. The lighting device of claim 9, comprising: at least one lighting source positioned on the base of the light assembly, wherein light emitted by the at least one lighting source positioned on the base of the light assembly is projected through said passthrough of said plurality of reflective surfaces.

12. The lighting device of claim 1, comprising: a filter positioned on a distal end of the lighting device, the filter configured to allow light within at least one wavelength range to pass through.

13. The lighting device of claim 1 comprising: an adjustment means configured to: alter the angular relation of the plurality of reflective surfaces.

14. A lighting device comprising: a lens assembly comprising: at least one lens, the at least one lens forming an optical axis associated with the lighting device; and a lighting assembly comprising: at least one first lighting source positioned on a base of the lighting assembly, the first lighting source configured to: emit light along the optical axis; at least one second lighting source positioned on an inner circumference of the lighting assembly, the second lighting source configured to: emit light along an axis substantially perpendicular to the optical axis; and a light director configured to: receive light emitted by the first lighting source and the second lighting source, wherein light emitted by the first lighting source is passed through the light director and light emitted by the second lighting source is re-directed by the light director to travel along the optical axis.

15. The lighting device of claim 14, comprising: an optical coating on the light director, the optical coating configured to: allow a desired wavelength range associated with the second light to be re-directed by the light director.

16. The lighting device of claim 14, wherein the optical coating is configured to: allow a desired wavelength range associated with the first light to be passed through the light director.

17. The lighting device of claim 14 comprising: a filter positioned on a distal end of the lighting device, the filter configured to: allow light within at least one wavelength range of the first light and second light to pass through.

18. A lighting device comprising: a lens assembly comprising: at least one lens element, the at least one lens element forming an optical axis through the lighting device; and a lighting assembly comprising: a first set of lighting sources arranged along an inner circumference of the lighting assembly; a second set of lighting sources arranged on a base of the lighting assembly; and a light director comprising: a plurality of reflective surfaces extending from the base at an acute angle toward an apex, the plurality of reflective surfaces forming a gap at the apex, wherein the light director is configured to: receive light emitted by the first set of lighting sources; re-direct the receive light toward a region on the at least one lens; and allow light emitted, along the optical axis, by the second set of lighting sources to pass through to the region on the at least one lens, wherein the light within the region after passing through the at least one lens converges onto a plane a

known distance from the first set of lighting sources.

19. The lighting device of claim 18, wherein the first set of lighting sources emit light along an axis substantially perpendicular to the optical axis.

20. The lighting device of claim 18, wherein the first set and second set of lighting sources comprise: at least one light emitting diode; and at least one of: an aperture holder, an aperture, and a dome lens.

21. The lighting device of claim 18, wherein the second set of lighting sources emit a white light and the first set of lighting sources emit light in a non-white wavelength band.
